

Equations with exact differentials. Integrating factor

A differential equation of order I, of the form

$$P(x, y) dx + Q(x, y) dy = 0, \quad (1.2.1)$$

with $(x, y) \in D$, D a *domain* from $\mathbb{R}^2$, is called **equation with exact differential** if there exists a function u , of two variables, such that

$$du = P(x, y) dx + Q(x, y) dy.$$

In this case, u can be determined from the relations $\frac{\partial u}{\partial x} = P(x, y)$ and $\frac{\partial u}{\partial y} = Q(x, y)$.

Theorem 1.2.1 *The equation (1.2.1) is with exact differential if and only if*

$$\frac{\partial P}{\partial y}(x, y) = \frac{\partial Q}{\partial x}(x, y), \quad \forall (x, y) \in D.$$

In this situation, a primitive u is given by the formula

$$u(x, y) = \int_{x_0}^x P(s, y) ds + \int_{y_0}^y Q(x_0, s) ds,$$

with $(x_0, y_0) \in D$ fixed, such that the rectangle $[x_0, x] \times [y_0, y]$ is included in D . The equation (1.2.1) is written in the form $du = 0$, so the general solution will be

$$u(x, y) = C, \quad C \in \mathbb{R}.$$

 **Example 1.2.1** Integrate the equation $(\cos xy - xy \sin xy) dx - x^2 \sin xy dy = 0$.

 **Solution:** We determine the partial derivatives of the functions P and Q :

$$\frac{\partial P}{\partial y}(x, y) = \frac{\partial Q}{\partial x}(x, y) = -2x \sin xy - x^2 y \cos xy$$

Since the derivatives are equal, it follows, according to Theorem 1.2.1, that the equation is with exact differential and we calculate

$$u(x, y) = \int_{x_0}^x (\cos sy - sy \sin sy) ds + \int_{y_0}^y -x_0^2 \sin x_0 s ds$$

Choosing $x_0 = y_0 = 0$, that belong to the domain of definition of the functions P and Q , we have

$$u(x, y) = \int_0^x (\cos sy - sy \sin sy) \, ds + \int_0^y 0 \, ds = x \cos xy.$$

So, the general solution of the equation is $x \cos xy = C$, $C \in \mathbb{R}$. ✓

In some cases, even if the equation is not with exact differential, there exists a nonconstant function $\mu(x, y)$, such that the equation

$$\mu(x, y)[P(x, y) \, dx + Q(x, y) \, dy] = 0$$

is with exact differential. Such a function is called **integrating factor**. Obviously, μ has to satisfy

$$\frac{\partial}{\partial y}(\mu P) = \frac{\partial}{\partial x}(\mu Q),$$

called **the equation of the integrating factor**. Generally, this is more difficult to solve than the initial equation. However, in certain situations this equation becomes quite simple:

1) If the expression $\frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right)$ depends only of the variable x , then the integrating factor is also a function only of x . It can be determined from the equation with separable variables

$$\mu'(x) = \frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) \mu(x). \quad (1.2.2)$$

Indeed, the equation of the integrating factor can be written:

$$\frac{\partial \mu}{\partial y} P + \mu \frac{\partial P}{\partial y} = \frac{\partial \mu}{\partial x} Q + \mu \frac{\partial Q}{\partial x}$$

and further on

$$\frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = \frac{1}{\mu} \left(-\frac{\partial \mu}{\partial y} \cdot \frac{P}{Q} + \frac{\partial \mu}{\partial x} \right).$$

If the left term of this relation depends only on x , it follows that we can search for μ depending only on x . We have that $\frac{\partial \mu}{\partial y} = 0$ and the equation (1.2.2) is obtained.

2) If $\frac{1}{P} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right)$ depends only on y , then μ also depends only on y and it can be determined from the equation

$$\mu'(y) = -\frac{1}{P} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) \mu(y).$$

Sometimes, we search for the integrating factor μ in the form $\mu = \mu(\varphi(x, y))$, with the function φ properly chosen.

 **Example 1.2.2** Solve the equation $y^2(x - y) dx + (1 - xy^2) dy = 0$, finding an integrating factor.

 **Solution:** We have $\frac{\partial P}{\partial y} = 2xy - 3y^2$ and $\frac{\partial Q}{\partial x} = -y^2$. The two partial derivatives are not equal, but $\frac{1}{P} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = \frac{2}{y}$ does not depend on x . Then we search for the integrating factor $\mu(y)$ as a solution of the equation:

$$\mu'(y) = -\frac{2}{y} \cdot \mu(y)$$

We have $\frac{\mu'}{\mu} = -\frac{2}{y}$, which implies $\mu(y) = \frac{1}{y^2}$. Multiplying the given equation with the integrating factor we obtain

$$(x - y) dx + \left(\frac{1}{y^2} - x \right) dy = 0,$$

that is with exact differential, and

$$u(x, y) = \int_{x_0}^x (s - y) ds + \int_{y_0}^y \left(\frac{1}{s^2} - x_0 \right) ds = \frac{x^2}{2} - xy - \frac{1}{y} + x_0 y_0 + \frac{1}{y_0} - \frac{x_0^2}{2}$$

The general solution is $\frac{x^2}{2} - xy - \frac{1}{y} = C$. 

 **Example 1.2.3** Solve the equation $(y + x^3 y^2) dx + (x + x^2 y^3) dy = 0$, knowing that it admits an integrating factor of the form $\mu = \mu(xy)$.

 **Solution:** Denote $P_1(x, y) = \mu(xy)(y + x^3 y^2)$, $Q_1(x, y) = \mu(xy)(x + x^2 y^3)$. From the condition $\frac{\partial P_1}{\partial y} = \frac{\partial Q_1}{\partial x}$ we obtain

$$\mu'(xy)x(y + x^3 y^2) + \mu(xy)(1 + 2x^3 y) = \mu'(xy)y(x + x^2 y^3) + \mu(xy)(1 + 2xy^3).$$

Denoting $xy = v$, after some calculations we get $\frac{\mu'}{\mu} = -\frac{2}{v}$, and from here

$$\mu = \frac{1}{v^2}, \text{ that is } \mu = \frac{1}{x^2 y^2}.$$

So we obtain the equation with exact differential

$$\left(\frac{1}{x^2 y} + x \right) dx + \left(\frac{1}{xy^2} + y \right) dy = 0,$$

with the general solution $\frac{x^2 + y^2}{2} - \frac{1}{xy} = C$. 



1.2.1 Integrate the equations:

i). $(2x + 3x^2y) dx + (x^3 - y^3) dy = 0$

ii). $e^{-y} dx - (2y + xe^{-y}) dy = 0$

iii). $\left(\frac{x}{\sin y} + 2 \right) dx + \frac{(x^2 + 1) \cos y}{\cos 2y - 1} dy = 0$

iv). $\frac{x dx + y dy}{\sqrt{x^2 + y^2}} + \frac{x dy - y dx}{x^2} = 0$

v). $\left(\frac{2 \cos 2x}{y} + \frac{1}{x} \right) dx - \left(\frac{\sin 2x}{y^2} + \frac{2 \sin y}{\cos^3 y} \right) dy = 0$

vi). $\left(3\sqrt{xy} - \frac{1}{x^2y} \right) dx + \left(\frac{x\sqrt{x}}{\sqrt{y}} - \frac{1}{xy^2} \right) dy = 0$

vii). $(3x^2y^2 - 4xy) dx + (2x^3y - 2x^2 + 3y^2) dy = 0, y(0) = 1$

viii). $\frac{2x}{y^3} dx + \frac{y^2 - 3x^2}{y^4} dy = 0, y(1) = 1$

1.2.2 Integrate the equations, finding the corresponding integrating factors:

i). $(x^2 + 3y) dx + 2x dy = 0$

ii). $3x^2y^2 dx + (y^4 \ln y - 2yx^3) dy = 0$

iii). $y dx - (4x^2y + x) dy = 0$

iv). $2x dx + (x^2 + y^2 + y) dy = 0$

v). $(xy + 1) dx + x(x + y) dy = 0$

vi). $(x \sin y + y \cos y) dx + (x \cos y - y \sin y) dy = 0$

vii). $\left(2 \cdot \frac{x}{y} - \frac{1}{x^2y^2} \right) dx + \frac{x^2}{y^2} dy = 0$

viii). $(1 - y\sqrt{y}) dx + \left(\frac{x}{2y} - 2x\sqrt{y} \right) dy = 0$

ix). $(y + 2x^2y) dx + x dy = 0$

x). $(\sin 2y + 2xy \cos^2 y) dx + (2x + x^2 \cos^2 y) dy = 0$

 **1.2.3** Integrate the equations, finding the integrating factors in the indicated form:

- i). $(x - y) dx + (x + y) dy = 0$, $\mu = \mu(x^2 + y^2)$
 ii). $(y^2 - x^2 + 2xy) dx + (y^2 - x^2 - 2xy) dy = 0$, $\mu = \mu(x^2 + y^2)$
 iii). $(xy \ln y + 2x^2y^2) dx + (x^2 + x^3y) dy = 0$, $\mu = \mu(xy)$
 iv). $(1 + x - e^{-x-y}) dx + (x - e^{-x-y}) dy = 0$, $\mu = \mu(x + y)$
 v). $(-xy \tan xy + y + 1) dx + (x - x^2 \tan xy) dy = 0$, $\mu = \mu(xy)$



Answers

1.2.1 i) $4x^2 + 4x^3y - y^4 = C$; ii) $xe^{-y} - y^2 = C$; iii) $\frac{x^2 + 1}{2 \sin y} + 2x = C$; iv) $\sqrt{x^2 + y^2} + \frac{y}{x} = C$; v) $\frac{\sin 2x}{y} + \ln x - \frac{1}{\cos^2 y} = C$; vi) $2x\sqrt{xy} + \frac{1}{xy} = C$; vii) $x^3y^2 - 2x^2y + y^3 = 1$; viii) $\frac{x^2}{y^3} - \frac{1}{y} = 0$.

1.2.2 i) $\mu(x) = x^{1/2}$, $x^{7/2} + 7yx^{3/2} = C$; ii) $\mu(y) = \frac{1}{y^4}$, $\frac{x^3}{y^2} + y \ln y - y = C$; iii) $\mu(x) = \frac{1}{x^2}$, $2y^2 + \frac{y}{x} = C$ iv) $\mu(y) = e^y$, $(x^2 + y^2 - y + 1)e^y = C$; v) $\mu(x) = \frac{1}{x}$, $xy + \frac{y^2}{2} + \ln |x| = C$; vi) $\mu(x) = e^x$, $e^x[(x - 1) \sin y + y \cos y] = C$; vii) $\mu(y) = y^2$, $x^2y + \frac{1}{x} = C$; viii) $\mu(y) = \sqrt{y}$, $x\sqrt{y} - xy^2 = C$; ix) $\mu(x) = e^{x^2}$, $xye^{x^2} = C$; x) $\mu(y) = \frac{1}{\cos^2 y}$, $2x \operatorname{tg} y + x^2y = C$.

1.2.3 i) $\mu(x, y) = \frac{1}{x^2 + y^2}$, $\ln(x^2 + y^2) - 2 \operatorname{arctg} \frac{y}{x} = C$; ii) $\mu(x, y) = \frac{1}{(x^2 + y^2)^2}$, $x - y = C(x^2 + y^2)$; iii) $\mu(x, y) = \frac{1}{xy}$, $x \ln y + x^2y = C$; iv) $\mu(x, y) = e^{x+y}$, $xe^{x+y} - x - y = C$; v) $\mu(x, y) = \cos xy$, $\sin xy + x \cos xy = C$.